Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 11)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how ionising radiation damages living cells.

[3 marks]

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0 1 . 2 Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

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0 1 . 3 Distinguish between **exposure** and **contamination**.

[2 marks]

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0 1 . 4 A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer ..... Sv

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0 1 . 5 State what is measured by the becquerel.

[1 mark]

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0 2 Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	**Metered households**
Mean water use / dm^3 per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 2 . 1 Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer ..... %

0 2 . 2 Explain why metering reduces water use.

[2 marks]

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0 2 . 3 Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

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0 2 . 4 Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 2 . 5 Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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0 3 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 3 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 3 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 4 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 4 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

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0 4 . 2 Explain why readings should be taken with the same wind direction.

[1 mark]

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0 4 . 3 Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

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0 4 . 4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

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0 5 Table 1 shows some techniques used to locate mineral deposits.

0 5 . 1 Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]

Table 1

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

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0 5 . 2 Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

0 5 . 3 Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

0 5 . 4 Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

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0 6 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO_2)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 6 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer ☐ tonnes

0 6 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer ☐ %

0 6 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO_2 concentrations.

[2 marks]

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0 6 . 4 Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

0 6 . 5 Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

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0 7 . 1 Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

0 7 . 2 Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

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0 7 . 3 Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

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0 7 . 4 Explain how high thermal mass building materials reduce energy use.

[2 marks]

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0 7 . 5 Describe **one** other energy storage technology.

[2 marks]

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0 8 A student investigated the effect of pH on the germination of cress seeds.

0 8 . 1 Explain why cress is a suitable species for this investigation but the results may not apply directly to natural ecosystems.

[2 marks]

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0 8 . 2 Describe how the student should decide the range of pH values to test.

[2 marks]

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0 8 . 3 Give **three** variables that should be standardised and explain how long the experiment should continue.

[4 marks]

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0 8 . 4 The student wants to compare the effect of nutrient concentration on the growth of duckweed. Explain why leaf area is a better measure of growth than fresh mass.

[2 marks]

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0 9 . 1 Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing .....

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Reducing soil biota .....

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0 9 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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0 9 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

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1 0 . 1 Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

[9 marks]

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Extra space

1 0 . 2 State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

1 1 . 2 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS